

CITYTWIN · REQUIREMENTS

CityTwin requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/citytwin		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from CTW-CAL-001; R12 and R17 redefined and R20 added per CTW-DDR-001; R1 restated for the pull paths (D11); reference rates from the sibling TRL 3 notes
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

CityTwin requirements

Targets remain proposals for review. The adopted items in CTW-DDR-001 (adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review) change three rows: R12 now covers the pilot on a sheltered site (D3), the outdoor range moves to a new R20, and R17 becomes a reported figure because the gateway is costed in TwinKit (D1). The status column comes from the TRL 3 calculation note CTW-CAL-001. Four requirements are **not met**: R10 (button response), R16 (kiosk cost against `budget_usd`), R18 (backup in the worst case) and R20 (street climate). R1 and R12 are **at risk**.

Table 1. Requirements.

ID	Requirement	Target	Status (CTW-CAL-001)	Verification (TRL 4 or later)
R1	Ingest every node type in the smart city set	All 9: seven over LoRaWAN through TwinKit; PotholeLog segment files and DockHub hourly aggregates pulled by outbound HTTPS from the operators' servers (D11)	At risk: six LoRaWAN types met on paper; the pull paths are proposed but not agreed with PotholeLog and DockHub; CrossSafe has no LoRaWAN uplink defined	Interface agreement with each sibling repo
R2	Capacity on one TwinKit gateway	50 nodes or more and 10,000 records per day or more	Met on paper: 46 nodes send 5,616 uplinks a day with 0.61 % lost (collisions and downlink blanking); 10,000 a day lose 0.71 %	Packet capture on a live gateway
R3	Accept only counts and levels	Each node type has a payload schema; any field outside it is rejected and logged; no images or audio stored	Met by design	Schema review; fuzz test of the decoder
R4	Suppress small counts in public outputs	People and vehicle counts below 5 per hour (D6) published as "fewer than 5"	Met by design	Review of export code
R5	No movement traces in public outputs	PotholeLog published per road segment only, no vehicle tracks or times finer than one day; DockHub per dock per hour, no card IDs	Met by design	Review of export code
R6	Open data export	Hourly CSV and GeoJSON with a documented schema, under CC BY 4.0 (D7), pushed to a public host	Met by design: 7,128 rows, 0.86 MB of CSV a day	Sample export checked against the schema
R7	One map for all node types	One layer per node type, node health, and an "as of" timestamp on every view	Met by design	Review of dashboard mock-up
R8	Data freshness	Map and kiosk show data no older than 30 min in normal operation	Met on paper, thin margin: 25.9 min worst on the kiosk, 15.6 min on the web map	Timing log on a live system
R9	Kiosk readable day and night	Readable in direct sun and at night at 1.5 m; key text cap height 7 mm or more	Met on paper: 41 px and 16.0 arcmin per capital; about 1,459 lx from the hood light	Mock-up viewed in sun and at night
R10	Kiosk responds to a button	Selected layer shown within 5 s of a press	Not met: 19.8 s; the button LED ring acknowledges the press at once	Timed on a bench build
R11	Accessible kiosk	Operable parts 380 to 1,220 mm above ground (ADA forward reach); overhang from the post no more than 305 mm between 685 and 2,030 mm (protruding objects); same content on a web page that meets	Met on paper: buttons at 1,120 mm; overhang 200 mm; web page not verifiable at TRL 3	Accessibility audit of the web page and a site review

ID	Requirement	Target	Status (CTW-CAL-001)	Verification (TRL 4 or later)
		WCAG 2.1 AA, linked by QR code and short URL		
R12	Pilot kiosk survives its climate (redefined, D3)	Sheltered or indoor site with no direct sun on the window; operate at 0 to 40 °C air; head IP54, cabinet IP55; panel within its 0 to 40 °C rating and the TwinKit pack within its 0 to 45 °C charge window	At risk: at 40 °C air the panel is at 40.1 °C and the pack at 44.4 °C (0.6 K margin)	Temperature log in a climate chamber
R13	Kiosk stands up to wind	35 m/s gust with a factor of 1.5 without yield of post or anchors	Met on paper: post 26.2 MPa, 1.92 kN per anchor	Structural design to the local code
R14	Electrical safety	Mains only inside the locked cabinet, behind a 30 mA RCBO and a surge protector; only 12 V SELV in the head; all metal earthed	Met by design; 12 V peak 26.7 W on a 60 W supply	Inspection by a qualified electrician
R15	Low running power	15 W average or less including the TwinKit gateway	Met on paper: 10.2 W (90 kWh a year)	Energy meter on a bench build
R16	Kiosk parts cost	Kiosk (items 1 to 12, 14, 15) within budget_usd of \$600; \$750 recommended under D1, awaiting Amish	Not met against \$600: \$739.00 (\$11.00 under the recommended \$750)	Priced BOM
R17	Cost with gateway (redefined, D1)	Kiosk plus TwinKit gateway reported; the gateway is costed and budgeted in the TwinKit repo	Met on paper (reported): \$1,029.00	Priced BOM
R18	Honest in an outage	Screen keeps the last image with its timestamp when power fails; gateway rides through 2 h on its backup pack	Not met in the worst case: 2.40 h with a new pack, 1.63 h aged at 0 °C (TWK-CAL-001); the e-paper keeps its image by design	Outage test
R19	Secure by default	No inbound connections from the internet to the gateway; open data pushed and sibling data pulled by outbound connections only; no exposed ports or USB on the kiosk	Met by design	Security review
R20	Street kiosk climate (new, target after the pilot, D3)	Operate from -20 °C to +45 °C air in full sun	Not met: panel about 79 °C in low sun at 45 °C air; a heated, insulated panel bay needs 13.5 W at -20 °C; cabinet pack out of its charge window in sun and in frost	Chamber and field tests

■ Assumptions

- Node types and reporting rates follow the sibling TRL 3 calculation notes: CurbCount, HeatMap Node, FloodGauge and NoiseMap 96 uplinks a day; AirStreet 288; LoadZone 124 (100 state changes and 24 heartbeats). CrossSafe's hourly activation summary is a CityTwin assumption; CrossSafe has not defined a LoRaWAN uplink.

- Radio loss is calculated (CTW-CAL-001 section A) rather than assumed; the TRL 2 figure of about 3 % was a placeholder.
- The pilot kiosk (R12) stands on a sheltered site; the street range (R20) is the target after the pilot.
- 35 m/s design gust is an estimate for a first check; the local wind code governs any real installation.